

Supporting online material

Materials and Methods

Study 1*Procedure and behavioral data*

We scanned participants in two groups of 12 each. Group 1 was composed largely of undergraduate students from Princeton University. Upon debriefing, it became apparent that a few of these participants were suspicious of the placebo manipulation because they had become suspicious of psychology experiments in general. Although our experiment wasn't advertised as a psychology experiment, the students knew that the scanner was housed in the Psychology building and therefore suspected that it was a Psychology experiment. To remedy this problem, participants for our second group of 12 were restricted to non-students who were unfamiliar with the Princeton University campus. Separate analyses for our two groups of 12 participants revealed no overall placebo effect for our student group (mean = 0.08, SD=0.54, $t(11) = 0.49$) and a significant effect for our non-student group (mean = 0.35, SD=0.37, $t(11) = 3.25$, $p < 0.01$). Informed consent was obtained from all participants after the nature and possible consequences of the study were explained.

We determined the current settings for intense and mild shocks on a participant-by-participant basis using the following procedure. With participants on the scanner gantry but not yet inside the magnet bore, a shock of 0.5 mA was applied and participants were asked 2 questions: 1) was that shock painful? 2) Can you tolerate a stronger shock? Shocks were repeated, raising the intensity by 0.5 mA each time, until participants answered “no” to the second question. Mild shock intensity was defined as the level of

the shock just prior to the point at which participants acknowledged pain (mean = 1.44 mA, sd = 0.85 mA). Intense shocks were set at the maximum level participants could tolerate (mean = 3.75 mA, sd = 2.34 mA).

The head-dedicated fMRI scanner has a short bore that made it possible to access the participants' wrist from outside the bore, without moving the participants. The participant could not see his or her hand as the investigator applied the skin cream. The cream was applied around the electrode site and massaged into the skin. The short bore also prevented the current in the wires from distorting the magnetic field and creating image artifacts. Participants were told that the cream required from 3-5 minutes before it would reach its full effectiveness, and that it was effective for only about 20 minutes (the approximate duration of the two scanning blocks in the placebo condition). In the control condition, participants were told that the skin cream would not affect their level of pain perception.

Half the cues were for intense shock and the other half were for mild shock, in random order. The 6 second shock interval consisted of a train of 5 electrical pulses of 20 ms duration at 1.2 second intervals.

fMRI acquisition and analysis

Anatomical scans were collected using a T1-weighted MP-RAGE protocol (256 X 256 matrix, FOV = 256 mm, 128 1.33 mm sagittal slices). Functional images were acquired with a Siemens 3.0 T head-dedicated MRI scanner using T2*-weighted EPI

(TR=3000 ms, TE=22 ms, 64 x 64 matrix, FOV = 192 mm, 30 2.5-mm axial slices with 1.5 mm gap). Five functional runs (153 scans each) were collected.

Preprocessing and data analysis were performed using Statistical Parametric Mapping software implemented in Matlab (SPM99; Wellcome Department of Cognitive Neurology, London, UK). Images were slice-timing corrected to the middle slice (slice 16), motion-corrected using a 6-parameter rigid-body transformation, and then spatially normalized to the Montreal Neurological Institute (MNI) template by using a 12-parameter affine transformation followed by nonlinear warping using basis functions (1). Images were subsequently smoothed with a Gaussian kernel of 6 mm FWHM. High frequency noise was removed by applying a low-pass filter (i.e., temporal smoothing) with the shape of the HRF. Low frequency noise was removed with a high pass filter using default cutoff periods. One participant was excluded due to excessive head motion (greater than 1.5 mm translation or 1.5 degrees of rotation within any run).

A random effects, event-related statistical analysis was performed with SPM99 (2). First, a separate general linear model (GLM) was specified for each participant, with regressors for each of the following five conditions: no shock, mild shock, intense shock, anticipation of mild shock, and anticipation of intense shock. This was done for Blocks 2-5, yielding a 20-column design matrix. Block 1 was excluded from analysis, as participants received neither the placebo nor the control skin cream during this block. Two of the remaining blocks involved placebo treatment and two involved control treatment, counterbalanced across participants so that half received placebo first (placebo-placebo-control-control block order) and half received control first (control-control-placebo-placebo order). There were no effects of order on behavioral data, and

order was not analyzed further. To examine the effect of intense shock, we calculated the contrast between the parameter estimates for intense shock and no-shock regressors across all four blocks. We also compared intense shocks with mild shocks to identify “pain-responsive” regions whose activity was more specific to the painful aspects of the stimulus. Voxels identified as pain responsive in [intense shock – no shock] and [intense shock – mild shock] comparisons served as a mask of pain regions within which to test for placebo effects. To examine the effect of anticipation of intense shock, we calculated the contrast between the parameter estimate for anticipation of intense shock and zero across all four blocks.

To identify voxels where the effect of intense shock differed between placebo and control trials (i.e., control > placebo), we specified a contrast that subtracted the difference between intense shock and no shock in the placebo blocks from the same difference in control blocks (e.g., (Shock with control – No shock with control) – (Shock with placebo – No shock with placebo)). To identify voxels in which anticipation of intense shock differed between placebo and control blocks (placebo > control), we specified a contrast that subtracted the effect of anticipation of intense shock in the placebo condition from anticipation of intense shock in the control condition.

Individual contrast images and ROI contrast values were entered into a second-level analysis, using a separate one-sample t-test that asked where the mean contrast value of the 24 participants differed from zero. In another second level analysis, correlation coefficients were calculated between each participant’s contrast value and their experienced placebo score (i.e., pain ratings for control – pain ratings for placebo in intense shock). In both cases, the resulting statistical maps were thresholded at $p < 0.005$

with a 10 voxel spatial extent threshold within pre-defined masks of pain-responsive regions (for pain analyses) and within dorsolateral PFC and orbitofrontal cortex (for anticipation analyses).

Study 2

Procedure and Behavioral data

Participants ($n = 67$) were recruited from the Ann Arbor community and underwent an initial testing session in the Ann Arbor VA Hospital. Participants ($n = 17$) who showed inconsistent pain ratings in the calibration phase (more than 4 points difference on a 10-point scale to successive presentations of the same temperature) or other pain-processing abnormalities (e.g., increasing pain in the absence of stimulation) were excluded from further testing. Behavioral placebo effects were tested in the remaining group ($n = 50$). Those who reported decreased pain in the placebo condition ($n = 36$) were screened for fMRI participation. FMRI participants were right handed, healthy volunteers with no history of psychiatric or neurological conditions.

Participants were tested in four phases. In Phase 1 (application), five patches of skin were treated at the start of the session with placebo and control creams. Both creams consisted of a base of Vaseline. A small amount of scented oil was added to the placebo cream to enhance the expectation that it was an effective pain reliever. However, tests on an additional group of participants ($n = 15$) showed that a placebo effect of the same magnitude was found when exactly the same creams were used. Two regions on either the upper or lower left forearm (counterbalanced across participants) were treated with the placebo cream (participants were told that it was “Lidocaine, a highly effective pain-relieving medication”). Another two regions (either upper or lower forearm, depending

on the location of placebo treatment) and a central region to be used for calibration were treated with a control cream (participants were told that it was a “sensory control, to control for having a cream on your skin, that will not block pain”). The design was similar to that of Price et al. (3), except that separate regions of the skin were used for manipulation and test to avoid confounding physiological carry-over effects with the placebo test.

In Phase 2 (calibration), a thermode (MEDOC, Inc.) was placed in the central region between placebo and control treatments. Two repetitions of 3 temperatures (starting at 45, 47, and 49 degrees Celsius) were administered, and temperatures were adjusted and the test repeated as necessary to find pain levels 2, 5, and 8 for each participant on a 10-point scale (1 was “just painful”, 10 was “unbearable pain”). On all trials, a 20-s thermal stimulation (17 s plateau, 1.5 s ramp up / ramp down to baseline) was followed by a 40-s rest period. Temperatures were 45.4 degrees centigrade on average (sd = 1.1) for Level 2, 47.0 (sd = 0.9) for Level 5, and 48.1 (sd = 1.0) for Level 8. During this phase, we collected continuous measurements during pain (using a manual sliding scale with 100 points) to determine the timecourse of reported pain during stimulation. Participants were told that following this procedure, they would receive four blocks of trials at their Level 8, two on placebo regions and two on control regions.

In Phase 3 (manipulation), one placebo-treated area and one control-treated area were selected, and a block of six stimulation trials was administered on each. The order was counterbalanced across participants. Thus, half the participants received stimulation on the control skin region first, and the other half received stimulation on the placebo region first. Manipulation and test blocks were administered in control-placebo-control-

placebo or placebo-control-placebo-control order. Unbeknownst to participants, during Phase 3 the control region was stimulated at Level 8, and the placebo region was stimulated at Level 2. The purpose of this manipulation was to convince participants that the placebo was truly effective at relieving pain (3, 4). Imaging was conducted during Phase 3, but the data were only used to define pain-responsive pain regions (see below).

Phase 4 (test) consisted of two blocks, one placebo and one control (counterbalanced), each occurring on previously unstimulated skin patches. The imaging data used to compare placebo and control activation was collected during this phase. Phase 4 was perceived by the participants as a continuation of Phase 3, and participants believed they were experiencing stimulation at Level 8. In Phase 4, stimulation was at Level 5 in both blocks, following the design of Price et al. (3). During fMRI scanning, exactly the same stimulus timing parameters were used for both test blocks, to avoid any systematic confounds occurring from interactions between stimulus timing and the BOLD response. All other methodological details were the same as in the manipulation phase. For the purpose of correlation analyses relating the magnitude of experienced placebo effects with brain placebo effects, placebo scores were calculated as the average (control – placebo) difference in experienced pain for each participant across the initial behavioral session and the subsequent fMRI session (Figure S3).

On all manipulation (Phase 3) and test (Phase 4) trials, a warning cue appeared on the screen that consisted of the words “Get ready!” followed by a variable-length anticipatory delay. A 20-s stimulation (17 s plateau, 1.5 s ramp up / ramp down to baseline), ensued, followed by a variable-length delay, a 4 s period in which the participant was asked to rate the pain experienced on that trial on a 10-point scale, and

finally a rest period lasting the duration of the 80 s trial (approximately 50 s). Pilot testing showed that this interval was adequate to prevent physiological carry-over effects across trials. The anticipation interval ($\bar{x} = 9.77$ s, $\sigma = 6.04$ s) and the interval from the end of stimulation to collection of the reported pain rating ($\bar{x} = 6.82$ s, $\sigma = 4.18$ s) were varied, to ensure separability of fMRI activation estimates for the anticipation, heat, and pain rating periods. The spacing between events within the trial was optimized for separability of estimation of the BOLD response to each phase using a genetic algorithm (5). In the initial VA session, participants rated pain on a visual analogue scale (VAS). In the fMRI session, participants rated pain using a 10-button response unit attached to both hands (Psychology Software Tools).

fMRI acquisition and analysis

Images were acquired on a GE Signa 3T scanner. Spiral GRE functional images were acquired at TR = 1.5, TE = 20, Flip = 90, 64 x 64 matrix, 3.75 x 3.75 x 5 mm voxels, skip 0. Prior to analysis, images were corrected for differences in slice acquisition time and head movement using MCFLIRT (6). A high-resolution spoiled GRASS structural image was collected for each participant and was spatially normalized to the Montreal Neurologic Institute (MNI) using SPM99 (1). Spatial transformations were applied to all functional images, and functional images were spatially smoothed with a 9 mm FWHM isotropic Gaussian kernel prior to analysis.

Data were analyzed in a multiple regression framework using SPM99, with boxcar regressors (i.e., estimators of the mean response during a period of time) for all time periods, shifted forward 4 s in time to allow for hemodynamic response lag. No

global signal scaling was used (7). Modeled periods were 4-s long for early anticipation and response periods, 6-s for anticipation, and 10-s for pain periods. After stimulus offset, experienced pain levels returned to baseline over a 12-14 s period. The fMRI response to the stimulation period was divided into three 10-s periods that corresponded to rising, peak plateau, and late heat phases (residual pain after stimulus offset) in the continuous pain self-report estimates obtained during calibration. The maximum pairwise correlation between predictors after high-pass filtering at 100 s was 0.5, between late heat and report periods. Image registration parameters were included in the regression analysis to minimize residual artifacts due to head motion that may remain after rigid-body motion correction. The control condition for each period was the adjusted session mean after removing all effects of interest, which consisted of activation in the long (~50 s) inter-trial interval.

As in Study 1, to determine regions of interest (ROIs) for analysis of placebo effects, data from pain trials during the test phase were first analyzed for effects of pain, collapsing across placebo and control (placebo + control). To find regions that responded to pain overall, we compared pain vs. baseline (inter-trial interval) activity. To find pain-responsive regions, which showed a greater response to more extreme pain, we compared pain responses to Level 8 vs. Level 2 in the manipulation phase. Since Level 2 was administered with placebo (as part of the manipulation), part of the difference in activation in Level 8 – Level 2 may reflect a placebo effect. However, such effects appear to be small relative to the effects of decreasing the temperature (Figure S3A), so we proceeded to use this comparison to identify pain-responsive regions. We examined placebo effects during the test phase in pain – baseline and pain-responsive regions,

treating participants as a random effect. As in Study 1, we used a threshold of $p < .005$ with 10 voxels spatial extent within the restricted search space of pain regions.

No significant effects of administration order (control first or placebo first) were found in behavioral pain ratings, and order-related effects were removed from imaging data on a voxel-by-voxel basis prior to statistical analysis by fitting a covariate to the group data. Coordinates were localized in Talairach coordinate space implemented in the Talairach Daemon (<http://ric.uthscsa.edu/projects/talairachdaemon.html>). Talairach coordinates falling within each anatomical volume of interest were transformed into MNI space using a bilinear transformation (<http://www.mrc-cbu.cam.ac.uk/Imaging/>).

Movement control (n = 17)

In an additional fMRI scanning period following collection of pain data, seventeen participants were asked to alternate between rotating their left forearms continuously back and forth at approximately 30 degrees of hand rotation and open-eye fixation baseline. Movement and rest periods alternated every 16 s (a value chosen for optimal detection sensitivity based on simulations (5)), and were cued by the appearance of either the word “Move” or the word “Rest,” which remained on the screen for 1 s and was followed by a fixation cross. This movement – rest comparison allowed us to localize primary and secondary motor regions in our group of participants. Pain activations in parietal cortex were functionally localized relative to these movement activations, and were posterior and inferior to activations created by arm movement.

References for online material

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Table S1. Summary of placebo effects in pain-responsive regions

Pain-responsive Regions					Study 1		Study 2					
Region	Coordinates				Shock pain		Early heat		Peak heat		Late heat	
	x	y	z	V	Max. t	Max. r	Max. t	Max. r	Max. t	Max. r	Max. t	Max. r
Medial frontal regions												
Pregenua anterior cingulate	2	40	-6	21	-1.62	0.12	-0.71	0.28	0.94	0.20	-0.77	0.23
	0	40	8	29	-0.74	0.16	-0.29	0.35	0.49	0.10	-0.25	0.09
Medial frontal pole	-6	48	16	98	-1.28	0.40	0.09	0.31	0.60	0.25	-0.13	0.20
	0	54	-18	192	-0.70	0.34	1.54	0.14	1.52	0.25	0.89	0.30
Mid-cingulate	0	-4	50	2841	1.21	0.61	2.06	0.58	2.02	0.31	2.95	0.36
Insular and temporal regions												
Superior insula / SII												
Left insula / SII	-54	-8	10	1258	1.56	0.57	1.45	0.46	2.46	0.32	3.91	0.36
Right insula / SII	52	-12	14	1408	1.54	0.55	3.46	0.26	1.85	0.13	3.72	0.18
Inferior insula												
	-38	6	-16	32	-0.13	0.34	-0.32	0.28	-0.19	0.34	2.19	0.18
	40	-6	-12	23	-0.20	0.31	1.31	-0.03	0.92	-0.08	3.81	-0.17
Temporal pole												
	-32	4	-28	16	1.32	0.44	-0.38	0.19	-0.23	0.01	1.66	-0.40
	-60	-38	-8	303	0.62	0.48	1.56	0.03	1.40	0.04	0.77	0.00
Uncus	-6	-2	-18	25	1.25	0.47	-0.40	0.14	1.26	0.01	0.85	-0.18
Other cortical regions												
M1 / S1	-40	-22	58	474	1.24	0.32	2.21	0.35	1.78	0.19	1.44	-0.02
M1 / S1	28	-32	64	41	-1.28	0.35	-0.54	0.06	-1.00	-0.04	0.54	-0.36
Left anterior PFC	-24	44	28	29	-0.33	0.46	-1.71	0.42	-1.82	-0.04	-0.27	0.30
Occipital cortex	22	-102	-6	17	-0.39	0.31	1.09	0.11	-0.57	0.19	-0.16	0.41
Subcortical regions												
Midbrain	10	-26	-10	61	0.43	0.17	-0.38	0.01	0.90	-0.02	2.11	-0.18
Thalamus												
	-10	-32	-6	21	0.76	0.25	-0.41	0.11	1.50	0.07	2.47	-0.03
	-4	-12	14	761	1.06	0.53	0.81	0.36	1.51	0.24	2.89	0.25
Caudate nucleus	0	10	18	26	-2.02	0.19	-0.65	0.15	-0.29	-0.19	2.34	-0.27
Cerebellum												
	-16	-44	-44	44	0.06	0.44	1.08	0.03	1.33	0.23	1.54	0.07
	-20	-30	-40	91	1.27	0.31	1.11	0.01	2.25	0.38	2.65	0.36
	26	-68	-40	64	1.14	0.31	1.32	-0.11	1.83	-0.20	1.63	0.00
	-22	-72	-36	50	1.47	0.39	0.66	0.21	-0.40	0.23	1.54	0.09
	-8	-88	-34	32	0.29	0.26	0.02	0.58	0.86	0.36	-0.81	0.56
	4	-52	-22	2339	1.44	0.51	2.94	0.20	2.82	0.33	2.59	0.32

Table S1. Summary of pain-responsive pain regions and the statistics for maximal placebo effects within each region. Pain-responsive regions were defined as voxels

significant for [intense shock – mild shock] or [very painful heat – moderately painful heat] in either Study 1 or Study 2. Coordinates show the standard Montreal Neurological Institute locations for activation centers of mass. Voxels counts show the number of contiguous significant 2 x 2 x 2 mm voxels in the region. *V* denotes the number of contiguous voxels within the region (intense – mild pain) at $p < .005$. *Max. t* indicates the maximal t-value for placebo effects (control > placebo) in pain within the region. *Max. r* values indicate the maximum Pearson's correlation value in the region, where correlations are between brain placebo (control – placebo in brain during pain) and experienced placebo effects (control – placebo in report). Placebo effects were not considered significant unless they met the threshold of $p < .005$ with 10 contiguous voxels. However, the summary statistics in this table may be useful for showing regions in which the current study did not show placebo effects. For example, out of 474 pain-responsive voxels in left sensorimotor cortex, no voxel showed a larger t-value than 2.21 in either study (however, null findings must always be interpreted with caution).

Table S2. Placebo decreases in pain regions

Study 1: Shock	Coordinates			Voxel counts			
Region	x	y	z	IM regions	IB regions	r	t
No significant main effects (control > placebo)							
Correlations with placebo (control - placebo experienced correlated positively with control - placebo brain)							
Left (contralateral) insula	-44	14	-3	8	19	0.57	3.25
Rostral anterior cingulate	-2	32	19	5	16	0.60	3.53
Rostral anterior cingulate	4	23	27	3	27	0.61	3.58
Medial thalamus	11	-5	14	3	8	0.53	2.91
Placebo x intensity [(control intense shock - placebo intense shock) - (control mild shock - placebo mild shock)]							
Left (contralateral) insula	-39	2	-6	4	11	N/A	3.07
Study 2: Early Heat							
Main effects of placebo (control > placebo)							
Right (contralateral) SII	62	-38	20	19	26	N/A	3.46
Correlations with placebo (control - placebo experienced correlated positively with control - placebo brain)							
Cerebellum	-9	-88	-35	11	0	0.58	3.25
Rostral anterior cingulate	3	18	34	37	37	0.58	3.27
Left anterior frontal cortex	-35	40	30	0	157	0.75	5.21
Study 2: Peak Heat							
No significant main effects (control > placebo)							
No significant positive correlations							
Study 2: Late Heat							
Main effects of placebo (control > placebo)							
Right (contralateral) inferior insula	40	-5	-11	12	779	N/A	3.81
Right (contralateral) mid-insula	41	7	1	207	779	N/A	3.72
Left (ipsilateral) inferior insula	-40	-11	-9	12	259	N/A	3.28
Left (ipsilateral) SII	-60	-5	12	82	259	N/A	3.91
Medial thalamus	2	-15	9	10	138	N/A	2.89
Mid-cingulate	-4	-12	49	15	15	N/A	2.95
Correlations with placebo (control - placebo experienced positively correlated with control - placebo brain)							
Right superior frontal gyrus	15	21	60	0	14	0.55	3.05
Right cuneus	10	-77	30	0	13	0.55	3.03

Table S2. Pain regions showing significant decreases in activity with placebo. The column labeled IM Regions shows the number of contiguous placebo-responsive voxels in pain-responsive regions, defined by the [intense – mild] contrast, as in Table S1. The column labeled IB Regions lists the number of contiguous placebo-responsive voxels in regions responding to [Intense – Baseline] in either study. Thus, there were 207 contiguous voxels within pain-responsive regions showing significant placebo effects in right mid-insula in Study 2, and this activation was part of a cluster of 779 contiguous voxels within stimulation-responsive [Intense – Baseline] regions. For regions that showed significant positive correlations between brain placebo (control – placebo in brain) and experienced placebo effects (control – placebo in report), r values indicate the maximum Pearson's correlation value in the region. T indicates the maximum t-value for placebo effects (control > placebo or the correlation) in each region. N/A (not applicable) indicates that the region showed a main effect rather than a correlation. Significant correlations indicate that strong placebo responders showed reduced pain activation due to placebo in this region.

Table S3. Placebo-induced increases in a-priori regions in anticipation of pain

Region	Coordinates			voxels	Z
	x	y	z		
Study 1: Shock					
Correlations with placebo (control - placebo in reported pain correlated positively with placebo - control in neural activity)					
Right orbitofrontal cortex	38	38	-12	14	3.41
	24	30	-12	62	5.92
Left orbitofrontal cortex	-26	38	-12	32	3.44
Rostral anterior cingulate	6	14	26	40	4.23
	-4	26	26	21	3.68
Dorsal anterior cingulate	10	32	32	20	3.36
	-2	-8	24	68	3.50
	-8	6	36	64	3.54
Right dorsolateral prefrontal cortex	10	-14	40	19	3.25
	42	6	30	14	3.30
	52	18	28	42	3.30
Left dorsolateral prefrontal cortex	36	20	38	13	3.00
	-30	4	42	22	3.30
Midbrain*	10	-26	-14	1	2.64
Study 2: Heat					
Main effects of placebo (placebo - control in neural activity)					
Early anticipation (4-8 s post-cue)					
Right orbitofrontal cortex*	28	48	-10	1	2.58
Dorsal anterior cingulated	-16	-22	12	204	3.25
Right dorsolateral prefrontal cortex	42	4	30	55	2.79
Left dorsolateral prefrontal cortex	-42	14	30	100	3.34
Midbrain	-2	-26	-12	251	3.56
Late anticipation (8-12 s post-cue)					
Rostral anterior cingulated	10	16	20	79	2.91
Right premotor cortex	34	8	32	21	2.72
Dorsal anterior cingulated	-20	-20	24	38	2.87

Table S3. Placebo-induced increases (placebo > control) in a-priori control regions during the anticipation period for Study 1 and Study 2. Correlations (Study 1) are correlations between the magnitude of placebo effects in reported pain (control - placebo) and the magnitude of placebo-induced increases during anticipation (placebo - control).

* Regions with only one significant voxel do not meet our extent threshold (10 voxels),

but we include them because there was substantial evidence for larger activations in the other study.

Supplementary figure captions

Figure S1. Pain responses in Study 1 (green; shock stimulus delivered to R wrist) and Study 2 (red; heat stimulus delivered to L forearm, peak heat period). Areas of overlap between the two experiments are shown in yellow. The activation threshold is $p < .005$ and 10 voxels spatial extent. Thus, voxels shown in green served as a high-threshold mask within which to test for placebo effects in Study 1. Red voxels served as the pain mask in Study 2.

Figure S2. Anticipation of pain in Study 1 (green; shock stimulus delivered to R hand) and Study 2 (red; heat stimulus delivered to L forearm). Areas of overlap between the two experiments are shown in yellow. The activation threshold is $p < .005$ and 10 voxels spatial extent.

Figure S3. A) Behavioral placebo effects ($n = 50$) during an initial testing session in Study 2. Participants expected stimulation at a pain level of 8 on a 10-point scale during all conditions. During the manipulation phase (left bars), stimuli were presented at Level 2 (placebo-treated region) and Level 8 (control-treated region). During the test phase, stimuli were presented at Level 5 in both placebo and control conditions, although this value must be considered as only approximate because sensitization may occur over time. The placebo ratings were significantly lower than the control ratings in the test phase, $t(49) = 5.87$, $p < .0001$. Error bars represent standard errors of the mean. Additional

details are reported in Materials and Methods at Science Online. B) For Study 2, a scatterplot showing the magnitude of behavioral placebo effects (control – placebo) for each participant during the initial session (x-axis) and the fMRI session (y-axis). The correlation between placebo effects in the two tests was $r = 0.62$, $p < .05$.

Figure S4. A slice at $z = -16$ mm, through the orbitofrontal cortex, of the mean functional EPI image (A, Study 1) and spiral image (B, Study 2) from one participant in each study. Reduced signal in orbitofrontal cortex in Study 2 causes loss of power to detect orbitofrontal effects.

Figure S5. Scatterplots showing placebo effects in rostral anterior cingulate (rACC) effects in shock (Study 1, triangles with solid line) and early heat (Study 2, X's with dashed line). The x-axis shows experienced placebo effects (control – placebo), and the y-axis shows brain placebo effects in BOLD contrast in rACC (control – placebo). For display purposes, BOLD contrast values for each Study were normalized by the group standard deviation, to make the scales comparable across the different scanners and regression models used in each Study.

Figure S1

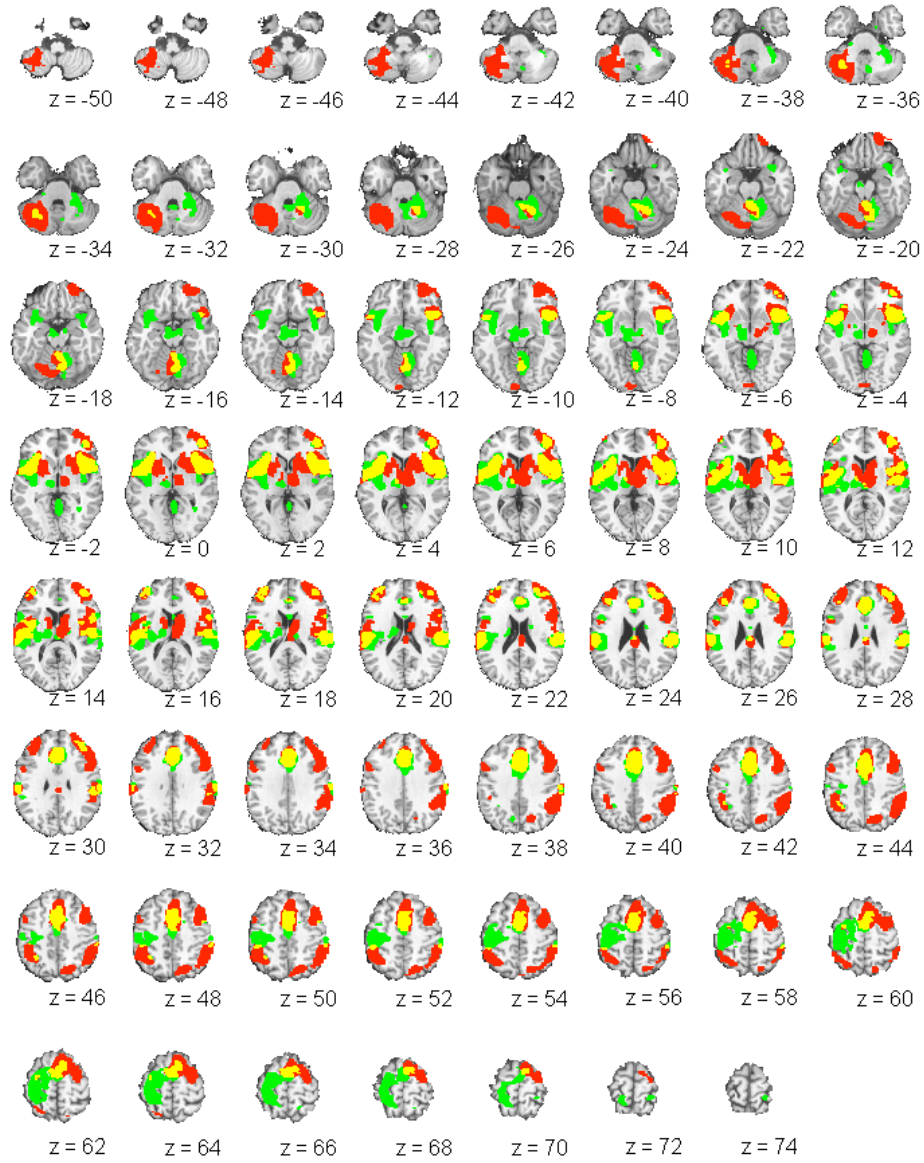


Figure S2.

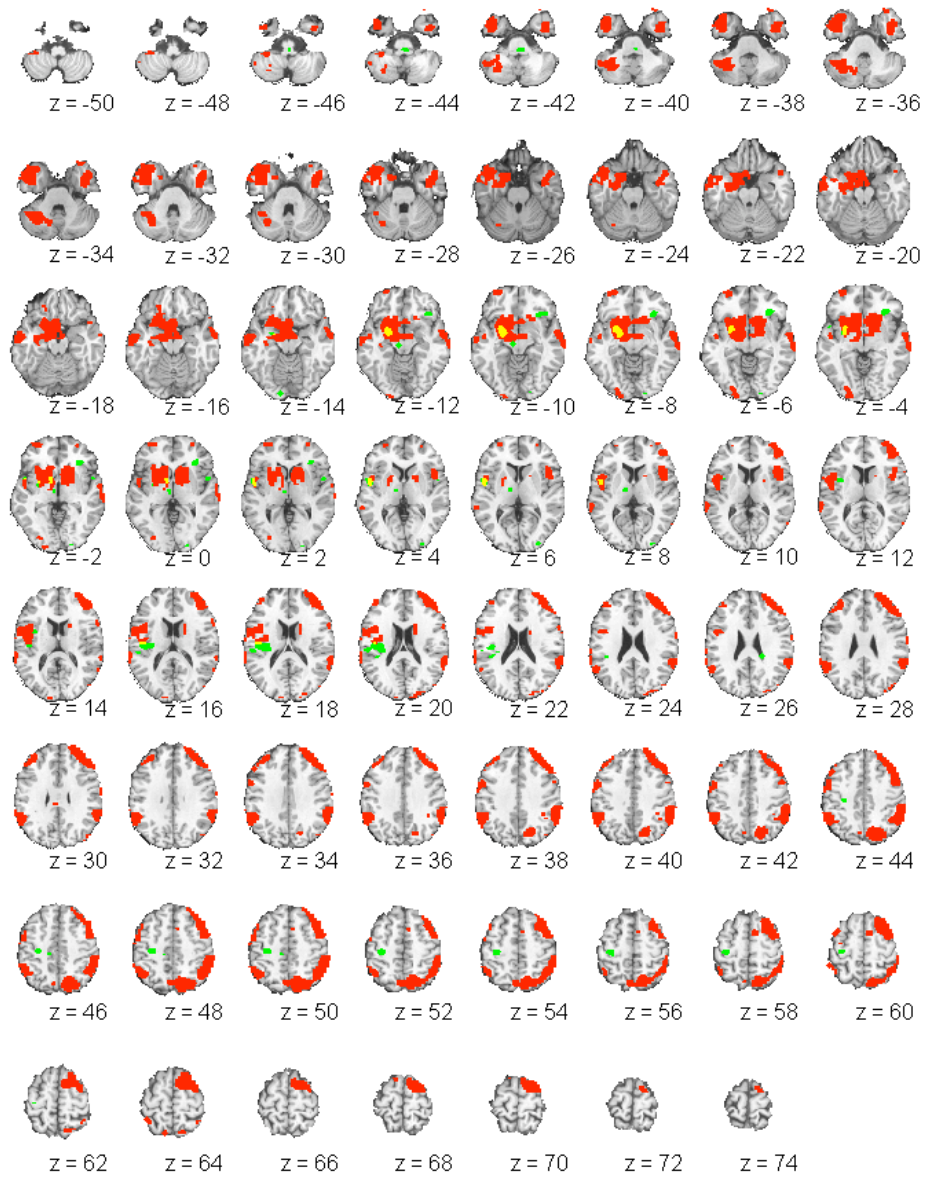


Figure S3.

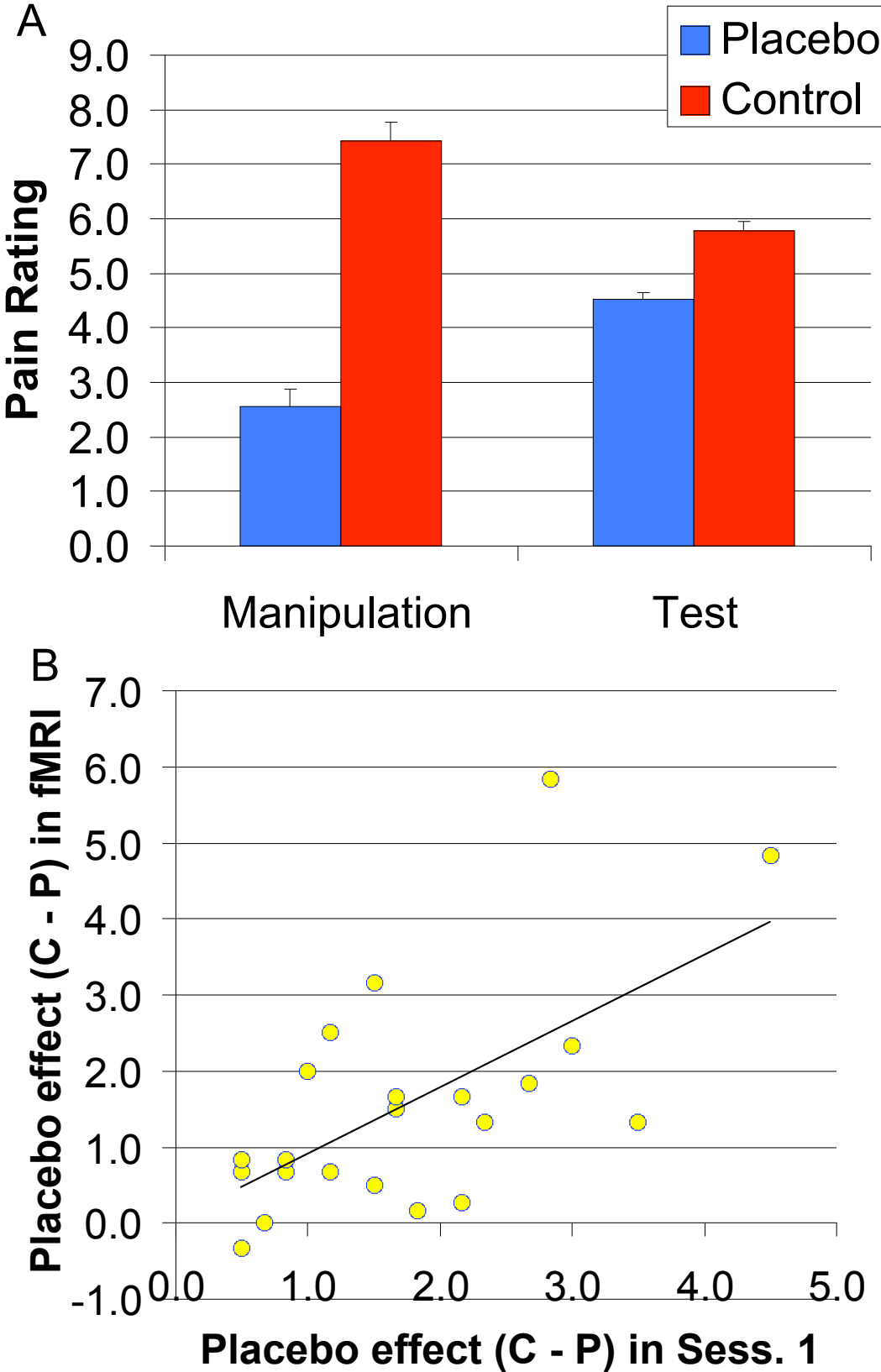


Figure S4.

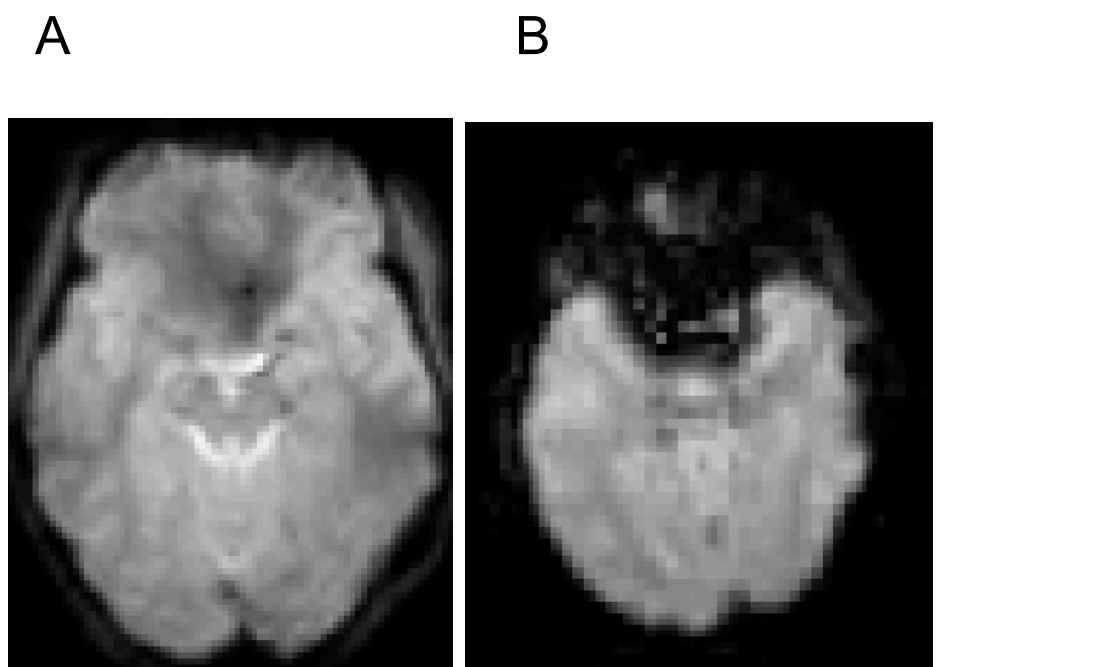


Figure S5.

